Prodigy Anesthesia Quick Review

Congenital Heart Disease

What type of shunt is associated with acyanotic congenital heart disease?	Left-to-right intracardiac shunt. The result of this shunt is pulmonary hypertension, right ventricular hypertrophy, and congestive heart failure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 49.*
When should surgical correction of an ASD be implemented?	ASDs should be surgically closed when pulmonary blood flow is 1.5 times the systemic blood flow. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
Describe the three different types of atrial septal defects (ASDs).	1. Ostium Primum (also called endocardial cushion defect) is a large opening located in the interatrial septum 2. Sinus venosus is a form of ASD found in the upper atrial septum 3. Ostium secundum (which is the most common type of ASD) is in the region of the fossa ovalis and ranges from a single opening to a septum that is fenestrated. They are frequently located close to the center of the interatrial septum. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 49-50.*
Regardless of where the shunt is located, what is the end result of acyanotic congenital heart disease?	Increased pulmonary blood flow with pulmonary hypertension, hypertrophy of the right ventricle, and after time, congestive heart failure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 49.*
Describe the medical treatment of PDAs.	Nonselective cyclooxygenase inhibitors, both COX-1 and COX-2 are used. These drugs inhibit prostaglandin synthesis. The use of indomethacin (a nonselective cyclooxygenase inhibitor) is the first line of therapy used for PDA, and has reduced the need for surgical closure by 60%. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 53.*
What risks are associated with the surgical closure of PDAs in infants?	Infections, intracranial hemorrhage, and paralysis of the recurrent laryngeal nerve *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 53.*
What type of murmur accompanies a patent ductus arteriosus?	A continuous systolic and diastolic murmur *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 52.*
What determines the pulmonary/systemic blood flow ratio in a patient with a patent ductus arteriosus?	1. The length and diameter of the ductus arteriosus 2. The pressure gradient that exists from the aorta to the pulmonary artery 3. The pulmonary/systemic vascular resistance ratio *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 52.*

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What is Eisenmenger's syndrome?	Eisenmenger's syndrome is a reversal of a left-to-right intracardiac shunt due to an increase in the pulmonary vascular resistance. Once the pulmonary vascular resistance reaches a level that is equal to or exceeds systemic vascular resistance, the shunt reverses to a right-to-left shunt. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 59.*
What is the principal indication for cardiac transplant?	Dilated cardiomyopathy is the primary indication for cardiac transplantation. Patients who respond the most favorably to this procedure are those younger than 60 years of age who were formerly otherwise healthy and whose symptoms are now worsening despite medical therapy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 139.*
What is the most common congenital cardiac abnormality seen in children and infants?	Ventricular septal defects (VSD) *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
What type of murmur is associated with aortic stenosis?	A systolic murmur. It can be heard over the second intercostal space on the right side and frequently radiates into the neck. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 53.*
List possible postoperative complications immediately following the surgical repair of coarctation of the aorta.	Paraplegia, hypertension, aortic regurgitation, infective endocarditis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 55.*
What are the clinical manifestations of a previously undetectable coarctation of the aorta?	Dizziness, palpitations, headache, and epistaxis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*
What is the hallmark symptom of cyanotic congenital heart disease?	Cyanotic congenital heart disease is comprised of a right-to-left intracardiac shunt and a decrease in blood flow to the pulmonary system, with resultant arterial hypoxemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
How do volatile anesthetics affect left-to-right shunts?	Volatile anesthetics cause a decrease in systemic vascular resistance, thus decreasing the left-to-right shunt. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What are the four anomalies associated with tetralogy of Fallot?	1. Hypertrophy of the right ventricle 2. VSD 3. An overriding aorta 4. Right ventricular outflow obstruction *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*

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Children with tetralogy of Fallot frequently assume a squatting position. Why is this?	Squatting causes an increase in systemic vascular resistance, creating an increase in pulmonary blood flow, and temporarily relieving their symptoms. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
Pregnant women with coarctation of the aorta are at an increased risk for what complication?	Aortic dissection *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*
List complications associated with coarctation of the aorta.	Aortic dissection, systemic hypertension, premature ischemic heart disease, left ventricular failure, CVAs resulting from the rupture of intracerebral aneurysms, and infective endocarditis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*
Distinguish the difference beween postductal and preductal coarctation of the aorta.	The two types of coarctation of the aorta are based upon the location of the narrowed area in relation to the position of the ductus arteriosus. Preductal coarctation of the aorta is located proximal to the opening of the ductus arteriosus, whereas in a postductal coarctation, the narrowing occurs distal to the ductus arteriosus. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 478.*
What type of murmur is associated with tetralogy of Fallot?	An ejection murmur caused from blood flowing across a stenotic pulmonary valve. This murmur can be auscultated along the sternal border on the left side. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
Describe congenital aortic stenosis.	Congenital aortic stenosis develops from a deformed bicuspid aortic valve. It is not stenotic at birth, but the valve leaflets thicken and calcify with time. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 53.*
How does an increase in pulmonary vascular resistance or a decrease in systemic vascular resistance affect the right-to-left intracardiac shunt in patients with tetralogy of Fallot?	The right-to-left shunt is increased, which increases arterial hypoxemia *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56-57.*
List the adverse side effects of indomethacin.	Indomethacin can result in renal and platelet dysfunction. *Cote CJ, Lerman J, & Anderson BJ. A Practice of Anesthesia for Infants and Children. 5th Ed. Philadelphia: Elsevier Saunders; 2013: 749.*

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What is a patent ductus arteriosus?	A patent ductus arteriosus is the failure of the ductus arteriosus to close after birth. Closure usually occurs within 24-48 hours after birth. The ductus arteriosus connects the descending aorta to the left pulmonary artery in the fetus, and it serves as a bypass for pulmonary arterial blood to directly enter the descending aorta rather than the deflated lungs. This failure to close results in a continuous blood flow from the aorta to the pulmonary artery. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 52.*
What is the effect of an increased systemic vascular resistance or decrease in pulmonary vascular resistance on a patient with an ASD?	There is an increase in the left-to-right shunt *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 49-50.*
Do patients with VSDs presenting for noncardiac surgery require prophylactic treatment against infective endocarditis?	Yes *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
What are the ECG manifestations of large VSDs?	Evidence of left ventricular and atrial enlargement is present. If pulmonary hypertension also exists, there is a rightward shift of the QRS axis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
Where would the murmur of a moderate to large VSD be auscultated the loudest?	The lower left sternal border. This murmur is holosystolic. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
What effect do large VSDs have on ventricular systolic pressures?	There is an equalization of ventricular systolic pressures. The degree of pulmonary and systemic blood flow is determined by the vascular resistances located within the two circulations. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
What are the two determinants of the physiological magnitude of a VSD?	1. The size of the VSD 2. The resistance in the pulmonary and systemic circulations *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*
Where are most VSDs located anatomically?	In the membranous portion of the intraventricular septum (70%). VSDs can also be found in the muscular portion of the septum (20%), close to the junction of the mitral and tricuspid valve (5%), and inferior to the aortic valve, resulting in aortic regurgitation (5%) *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 51.*

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How does positive-pressure ventilation affect left to right shunts?	Positive-pressure ventilation increases pulmonary vascular resistance, which results in a decrease in the left-to-right shunt. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What ECG findings are typical in patients with congenital aortic stenosis?	There is evidence of left ventricular hypertrophy, and ST segment depression often occurs during exercise. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54.*
What effect does the use of a high FiO2 have on a patient with an ASD?	There is a decrease in pulmonary vascular resistance, which results in an increase in pulmonary blood flow and left-to-right shunt. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What is the most common congenital abnormality that results in aortic dissection?	Although Marfan syndrome and Ehlers-Danlos syndrome are associated with an increased risk for aortic aneurysm and dissection, bicuspid aortic valve is far more common, occurring in 1% of the population. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 155.*
Why would a patient with a double aortic arch avoid flexion of their neck?	Flexion of the neck worsens tracheal compression. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67.*
What are some signs and symptoms of a double aortic arch?	Dysphagia, inspiratory stridor, and difficulty in the mobilization of secretions *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67.*
What is tricuspid atresia?	Tricuspid atresia is a type of congenital heart disease in which there is a complete absence of the tricuspid valve. The right ventricle is small, the left ventricle is enlarged, arterial hypoxemia is present, and there is a significant decrease in pulmonary blood flow. An ASD is present, and pulmonary blood flow occurs from a VSD, bronchial blood vessels, or a PDA. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 61.*
What is the most common cyanotic congenital heart disease?	Tetralogy of Fallot *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
When a patient with Eisenmenger's syndrome undergoes noncardiac surgery, what should the maintenance of anesthesia be based upon?	Maintaining the same degree of systemic vascular resistance that existed preoperatively will help avoid an increase in the right-to-left shunt that occurs with vasodilation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 59-60.*

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What impact does an increase in systemic vascular resistance have on a patient with an ASD?	There is an increase in the left-to-right shunt *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What are the symptoms associated with large ASDs?	Right heart failure, supraventricular dysrhythmias, dyspnea on exertion, recurrent pulmonary infections, paradoxical embolism *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What are signs and symptoms of Eisenmenger's syndrome?	As the right-to-left shunt increases, decreased exercise tolerance and cyanosis develops. Erythrocytosis develops as a result of arterial hypoxemia, which increases the viscosity of the blood, leading to headaches, dizziness, paresthesias, and visual problems. Palpitations often occur due to the onset of atrial flutter or atrial fibrillation. Hemoptysis and coagulopathies are seen, as well as an increased chance of CVA and brain abscess. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 59.*
What is Ebstein's anomaly?	A tricuspid valve abnormality in which the valve leaflets are displaced into the right ventricle, or the leaflets are malformed. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 60.*
List three things that can increase the right-to-left shunt of patients with TOF.	1. An increase in myocardial contractility 2. An increase in pulmonary vascular resistance 3. A decrease in systemic vascular resistance *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56-57.*
What is the key to understanding the management of anesthesia in patients with tetralogy of Fallot?	An in-depth knowledge of the drugs and different events affecting the extent of the right-to-left shunt. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
What is truncus arteriosus?	Truncus arteriosus is a congenital heart defect in which the aorta and pulmonary arteries derive from a single trunk that overrides both the left and right ventricles. A VSD is present, and the mortality rate is high. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 64.*
What are the signs and symptoms of truncus arteriosus?	Failure to thrive, congestive heart failure, arterial hypoxemia and cyanosis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 64.*

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Management of anesthesia in patients with truncus arteriosus is influenced by what main factor?	The degree of pulmonary blood flow that is present. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 64.*
Describe the difference between partial and total anomalous pulmonary venous return.	Partial anomalous pulmonary venous return is a cardiac defect in which the left or right pulmonary veins empty into the right side of circulation instead of the left atrium. In total anomalous pulmonary venous return, all four pulmonary veins drain into the systemic venous system. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 65.*
Should patients with PDAs presenting for noncardiac surgery be given prophylactic antibiotics to protect against infective endocarditis?	Yes *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 53.*
What is the significance of a double aortic arch?	A double aortic arch creates a vascular ring that produces compression of the trachea and esophagus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67.*
Without surgical correction, what is the mortality rate of patients with tetralogy of Fallot?	Over 50% by the age of 3 *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 57.*
What congenital heart disorders create mechanical obstruction to the trachea?	Aberrant left pulmonary artery, double aortic arch, absent pulmonic valve *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67-68.*
What are the signs and symptoms of an aberrant left pulmonary artery?	Expiratory stridor, arterial hypoxemia, wheezing *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67-68.*
What is an aberrant left pulmonary artery?	A congenital heart defect in which there is an absent left pulmonary artery. The left lung receives arterial blood via a branch of the right pulmonary artery that passes between the trachea and esophagus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 67-68.*
What is the result of an absent pulmonic valve?	Pulmonary artery dilation which can cause compression of the left mainstem bronchus and trachea. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 68.*

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Should patients undergoing dental or surgical procedures with tetralogy of Fallot receive antibiotics for prophylactic treatment of infective endocarditis?	Yes *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 57.*
What is hypoplastic left heart syndrome?	A congenital heart defect in which parts of the left side of the heart are not completely developed. The left ventricle, mitral valve, and ascending aorta are hypoplastic, aortic valve atresia is present, and there is a patent ductus arteriosus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 65-66.*
How is pulmonic stenosis identified in patients not exhibiting signs or symptoms of disease?	A loud systolic ejection murmur is present. It can be best heard on the left side at the second intercostal space. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54.*
What considerations should be given during the management of anesthesia during a surgical resection of coarctation of the aorta?	1. The likelihood of systemic hypertension once the aorta is cross-clamped 2. Adequacy of blood flow to the lower body once the aorta is cross-clamped 3. The risk of spinal cord ischemia and the neurologic outcomes that would occur due to the cross-clamping of the aorta. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 55.*
What mean arterial pressure is desired in the lower extremities during aortic cross-clamping to ensure blood flow is adequate to the spinal cord and kidneys?	40 mmHg or greater *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 55.*
Does a patient with an ASD require prophylactic treatment against infective endocarditis?	Unless there is an associated valvular disorder, patients with an ASD do not require prophylactic treatment against infective endocarditis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 50.*
What arrhythmias are common in patients following surgical repair of tetralogy of Fallot?	Atrial fibrillation, atrial flutter, right bundle branch blocks *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 59.*
What other disorders may accompany patients with coarctation of the aorta?	PDA, aneurysms of the Circle of Willis, mitral regurgitation or stenosis, Turner's syndrome, bicuspid aortic valve. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*

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Describe transposition of the great vessels.	In transposition of the great vessels, the aorta arises from the right ventricle, and the pulmonary artery arises from the left ventricle. The result is a complete separation of systemic and pulmonary circulations. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 62-63.*
When managing the anesthetic of a patient with congenital pulmonic stenosis, what cardiac changes must be avoided?	Increases in myocardial contractility and heart rate must be avoided due to the increased oxygen demand it places on the right ventricle. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54.*
At what point should surgical repair for coarctation of the aorta be considered?	When the transcoarctation pressure gradient is more than 30 mmHg, surgical intervention should be considered. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*
How is congenital pulmonic stenosis treated?	Percutaneous balloon valvuloplasty *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54.*
Describe the characteristics of hypercyanotic attacks.	Hypercyanotic attacks frequently accompany exercise or crying and are associated with tachypnea, worsening arterial hypoxemia and cyanosis. In extreme instances, seizures, loss of consciousness, cerebrovascular accidents, and even death can occur. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 57.*
What pharmacologic agents cause a decrease in systemic vascular resistance and result in an increase in the right-to-left shunt of patients with TOF?	Alpha adrenergic blockers, volatile anesthetics, ganglionic blockers, and agents that evoke histamine release. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 58-59.*
What intracranial disorder are patients with cyanotic congenital heart disease at major risk for developing?	A brain abscess *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 56.*
How does an increase in systemic vascular resistance or a decrease in pulmonary vascular resistance affect the left-to-right shunt in a patient with a PDA?	The left-to-right-shunt is increased *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 52-53.*

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How does a decrease in systemic vascular resistance affect the left-to-right shunt in a patient with a PDA?	The left-to-right shunt is decreased, therefore, systemic blood flow is improved. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 424.*
Is coarctation of the aorta seen more frequently in males or females?	Males *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 54-55.*

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